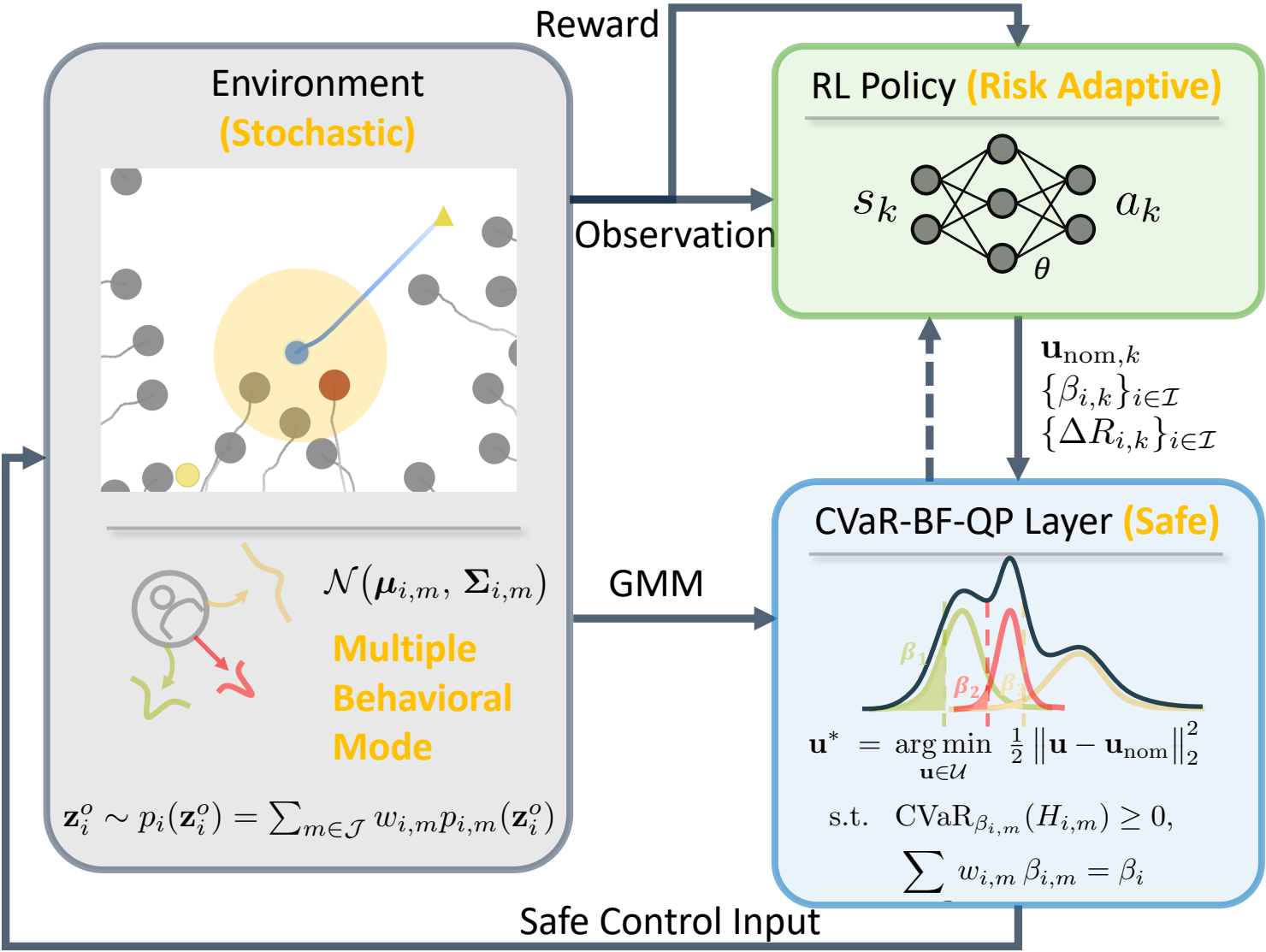
Reward
Environment
(Stochastic)
RL Policy (Risk Adaptive)
s_k
a_k
θ
Observation
$\mathbf{u}_{\text{nom},k}$
$\{\beta_{i,k}\}_{i\in\mathcal{I}}$
$\{\Delta R_{i,k}\}_{i\in\mathcal{I}}$
$\mathcal{N}(\boldsymbol{\mu}_{i,m}, \boldsymbol{\Sigma}_{i,m})$
Multiple Behavioral Mode
GMM
CVaR-BF-QP Layer (Safe)
β_1
β_2
β_3
$\mathbf{u}^* = \arg\min_{\mathbf{u}\in\mathcal{U}} \frac{1}{2}\|\mathbf{u}-\mathbf{u}_{\text{nom}}\|_2^2$
s.t. $\text{CVaR}_{\beta_{i,m}}(H_{i,m}) \geq 0,$
$\sum w_{i,m}\beta_{i,m} = \beta_i$
$\mathbf{z}_i^o \sim p_i(\mathbf{z}_i^o) = \sum_{m\in\mathcal{J}} w_{i,m} p_{i,m}(\mathbf{z}_i^o)$
Safe Control Input